

2.3.2 Memory map and register boundary addresses

Table 7. Memory map and default device memory area attributes

Region	Boundary address	Arm® Cortex®-M7	Type	Attributes	Execute never
External Devices	0xD0000000 - 0xDFFFFFFF	FMC SDRAM Bank2 (or Reserved in case of FMC remap)	Device	-	Yes
	0xCC000000 - 0xCFFFFFFF	FMC SDRAM Bank1 (or remap of FMC NOR/PSRAM/SRAM 4 Bank1)			
	0xC8000000 - 0xCBFFFFFF	FMC SDRAM Bank1 (or remap of FMC NOR/PSRAM/SRAM 3 Bank1)			
	0xC4000000 - 0xC7FFFFFF	FMC SDRAM Bank1 (or remap of FMC NOR/PSRAM/SRAM 2 Bank1)			
	0xC0000000 - 0xC3FFFFFF	FMC SDRAM Bank1 (or remap of FMC NOR/PSRAM/SRAM 1 Bank1)			
	0xA0000000 - 0xBFFFFFFF	Reserved			
External Memories	0x90000000 - 0x9FFFFFFF	QUADSPI	Normal	Write-through cache attribute	No
	0x80000000 - 0x8FFFFFFF	FMC NAND Flash memory		Write-back, write allocate cache attribute	
	0x70000000 - 0x7FFFFFFF	Reserved (or remap of FMC SDRAM Bank2)			
	0x6C000000 - 0x6FFFFFFF	FMC NOR/PSRAM/SRAM 4 Bank1 (or remap of FMC SDRAM Bank1)			
	0x68000000 - 0x6BFFFFFF	FMC NOR/PSRAM/SRAM 3 Bank1 (or remap of FMC SDRAM Bank1)			
	0x64000000 - 0x67FFFFFF	FMC NOR/PSRAM/SRAM 2 Bank1 (or remap of FMC SDRAM Bank1)			
	0x60000000 - 0x63FFFFFF	FMC NOR/PSRAM/SRAM 1 Bank1 (or remap of FMC SDRAM Bank1)			
Peripherals	0x40000000 - 0x5FFFFFFF	Peripherals (refer to <i>Table 8: Register boundary addresses</i>)	Device	-	Yes

Table 7. Memory map and default device memory area attributes

Region	Boundary address	Arm® Cortex®-M7	Type	Attributes	Execute never
RAM	0x38801000 - 0x3FFFFFFF	Reserved	Normal	Write-back, write allocate cache attribute	No
	0x38800000 - 0x38800FFF	Backup SRAM			
	0x38010000 - 0x387FFFFF	Reserved			
	0x38000000 - 0x3800FFFF	SRAM4			
	0x30048000 - 0x37FFFFFF	Reserved			
	0x30040000 - 0x30047FFF	SRAM3			
	0x30020000 - 0x3003FFFF	SRAM2			
	0x30000000 - 0x3001FFFF	SRAM1			
	0x24080000 - 0x2FFFFFFF	Reserved			
	0x24000000 - 0x2407FFFF	AXI SRAM			
	0x20020000 - 0x23FFFFFF	Reserved			
	0x20000000 - 0x2001FFFF	DTCM			
Code	0x1FF20000 - 0x1FFFFFFF	Reserved	Normal	Write-through cache attribute	No
	0x1FF00000 - 0x1FF1FFFF	System Memory			
	0x08200000 - 0x1FEFFFFF	Reserved			
	0x08100000 - 0x081FFFFF	Flash memory bank 2 ⁽¹⁾			
	0x08000000 - 0x080FFFFF	Flash memory bank 1 ⁽²⁾			
	0x00010000 - 0x07FFFFFF	Reserved			
	0x00000000 - 0x0000FFFF	ITCM			

- Flash memory bank 2 boundary is limited to 0x08100000 - 0x0817FFFF on STM32H742xG/STM32H743xG. It is reserved on STM32H750xB.
- Flash memory bank 1 boundary is limited to 0x08000000 - 0x0807FFFF and 0x08000000 - 0x0801FFFF on STM32H742xG/STM32H743xG and STM32H750xB boundary, respectively.

All the memory map areas that are not allocated to on-chip memories and peripherals are considered “Reserved”. For the detailed mapping of available memory and register areas, refer to the following table.

The following table gives the boundary addresses of the peripherals available in the devices.

Table 8. Register boundary addresses⁽¹⁾

Boundary address	Peripheral	Bus	Register map
0x58027000 - 0x580273FF	RAMECC3		Section 3.4: RAMECC registers
0x58026400 - 0x580267FF	HSEM	AHB4 (D3)	Section 10.4: HSEM registers
0x58026000 - 0x580263FF	ADC3		Section 25.7: ADC common registers
0x58025800 - 0x58025BFF	DMAMUX2		Section 17.6: DMAMUX registers
0x58025400 - 0x580257FF	BDMA		Section 16.6: BDMA registers
0x58024C00 - 0x58024FFF	CRC		Section 21.4: CRC registers
0x58024800 - 0x58024BFF	PWR		Section 6.8: PWR register description
0x58024400 - 0x580247FF	RCC		Section 8.7: RCC register description
0x58022800 - 0x58022BFF	GPIOK		Section 11.4: GPIO registers
0x58022400 - 0x580227FF	GPIOK		Section 11.4: GPIO registers
0x58022000 - 0x580223FF	GPIOK		Section 11.4: GPIO registers
0x58021C00 - 0x58021FFF	GPIOK		Section 11.4: GPIO registers
0x58021800 - 0x58021BFF	GPIOK		Section 11.4: GPIO registers
0x58021400 - 0x580217FF	GPIOK		Section 11.4: GPIO registers
0x58021000 - 0x580213FF	GPIOK		Section 11.4: GPIO registers
0x58020C00 - 0x58020FFF	GPIOD		Section 11.4: GPIO registers
0x58020800 - 0x58020BFF	GPIOD		Section 11.4: GPIO registers
0x58020400 - 0x580207FF	GPIOD		Section 11.4: GPIO registers
0x58020000 - 0x580203FF	GPIOD		Section 11.4: GPIO registers

Table 8. Register boundary addresses⁽¹⁾ (continued)

Boundary address	Peripheral	Bus	Register map
0x58005800 - 0x58006BFF	Reserved	APB4 (D3)	Reserved
0x58005400 - 0x580057FF	SAI4		Section 51.6: SAI registers
0x58004C00 - 0x58004FFF	Reserved		Reserved
0x58004800 - 0x58004BFF	IWDG1		Section 45.4: IWDG registers
0x58004000 - 0x580043FF	RTC & BKP registers		Section 46.6: RTC registers
0x58003C00 - 0x58003FFF	VREF		Section 27.3: VREFBUF registers
0x58003800 - 0x58003BFF	COMP1 - COMP2		Section 28.7: COMP registers
0x58003000 - 0x580033FF	LPTIM5		Section 43.7: LPTIM registers
0x58002C00 - 0x58002FFF	LPTIM4		Section 43.7: LPTIM registers
0x58002800 - 0x58002BFF	LPTIM3		Section 43.7: LPTIM registers
0x58002400 - 0x580027FF	LPTIM2		Section 43.7: LPTIM registers
0x58001C00 - 0x58001FFF	I2C4		Section 47.7: I2C registers
0x58001400 - 0x580017FF	SPI6		Section 50.11: SPI/I2S registers
0x58000C00 - 0x58000FFF	LPUART1		Section 49.6: LPUART registers
0x58000400 - 0x580007FF	SYSCFG		Section 12.3: SYSCFG registers
0x58000000 - 0x580003FF	EXTI		Section 20.6: EXTI registers
0x52009000 - 0x520093FF	RAMECC1		Section 3.4: RAMECC registers
0x52008000 - 0x52008FFF	Delay Block SDMMC1	AHB3 (D1)	Section 24.4: DLYB registers
0x52007000 - 0x52007FFF	SDMMC1		Section 55.9: SDMMC registers
0x52006000 - 0x52006FFF	Delay Block QUADSPI		Section 24.4: DLYB registers
0x52005000 - 0x52005FFF	QUADSPI control registers		Section 23.5: QUADSPI registers
0x52004000 - 0x52004FFF	FMC control registers		Section 22.7.6: NOR/PSRAM controller registers , Section 22.8.7: NAND Flash controller registers , Section 22.9.5: SDRAM controller registers
0x52003000 - 0x52003FFF	JPEG		Section 33.5: JPEG codec registers
0x52002000 - 0x52002FFF	Flash interface registers		Section 4.9: FLASH registers
0x52001000 - 0x52001FFF	Chrom-Art (DMA2D)		Section 18.6: DMA2D registers
0x52000000 - 0x52000FFF	MDMA		Section 14.5: MDMA registers
0x51000000 - 0x510FFFFF	GPV		Section 2.2: AXI interconnect matrix (AXIM)
0x50003000 - 0x50003FFF	WWDG1	APB3 (D1)	Section 44.5: WWDG registers
0x50001000 - 0x50001FFF	LTDC		Section 32.7: LTDC registers
0x50000000 - 0x50000FFF	Reserved		-
0x48023000 - 0x480233FF	RAMECC2		Section 3.4: RAMECC registers

Table 8. Register boundary addresses⁽¹⁾ (continued)

Boundary address	Peripheral	Bus	Register map
0x48022800 - 0x48022BFF	Delay Block SDMMC2	AHB2 (D2)	Section 24.4: DLYB registers
0x48022400 - 0x480227FF	SDMMC2		Section 55.9: SDMMC registers
0x48021800 - 0x48021BFF	RNG		Section 34.7: RNG registers
0x48021400 - 0x480217FF	HASH		Section 36.7: HASH registers
0x48021000 - 0x480213FF	CRYPTO		Section 35.6: CRYPT registers
0x48020000 - 0x480203FF	DCMI		Section 31.5: DCMI registers
0x40080000 - 0x400BFFFF	USB2 OTG_FS	AHB1 (D2)	Section 57.14: OTG_HS registers
0x40040000 - 0x4007FFFF	USB1 OTG_HS		Section 57.14: OTG_HS registers
0x40028000 - 0x400293FF	ETHERNET MAC		Section 58.11: Ethernet registers
0x40024400 - 0x400247FF	Reserved		Reserved
0x40022000 - 0x400223FF	ADC1 - ADC2		Section 25.7: ADC common registers
0x40020800 - 0x40020BFF	DMAMUX1		Section 17.6: DMAMUX registers
0x40020400 - 0x400207FF	DMA2		Section 15.5: DMA registers
0x40020000 - 0x400203FF	DMA1		Section 15.5: DMA registers
0x40017400 - 0x400177FF	HRTIM	APB2 (D2)	Section 37.5: HRTIM registers
0x40017000 - 0x400173FF	DFSDM1		Section 30.7: DFSDM channel y registers (y=0..7), Section 30.8: DFSDM filter x module registers (x=0..3)
0x40016000 - 0x400163FF	SAI3		Section 51.6: SAI registers
0x40015C00 - 0x40015FFF	SAI2		Section 51.6: SAI registers
0x40015800 - 0x40015BFF	SAI1		Section 51.6: SAI registers
0x40015000 - 0x400153FF	SPI5		Section 50.11: SPI/I2S registers
0x40014800 - 0x40014BFF	TIM17		Section 41.6: TIM16/TIM17 registers
0x40014400 - 0x400147FF	TIM16		Section 41.6: TIM16/TIM17 registers
0x40014000 - 0x400143FF	TIM15		Section 41.5: TIM15 registers
0x40013400 - 0x400137FF	SPI4		Section 50.11: SPI/I2S registers
0x40013000 - 0x400133FF	SPI1 / I2S1		Section 50.11: SPI/I2S registers
0x40011400 - 0x400117FF	USART6		Section 48.7: USART registers
0x40011000 - 0x400113FF	USART1		Section 48.7: USART registers
0x40010400 - 0x400107FF	TIM8		Section 38.4: TIM1/TIM8 registers
0x40010000 - 0x400103FF	TIM1		Section 38.4: TIM1/TIM8 registers

Table 8. Register boundary addresses⁽¹⁾ (continued)

Boundary address	Peripheral	Bus	Register map
0x4000AC00 - 0x4000D3FF	CAN Message RAM	APB1 (D2)	Section 56.5: FDCAN registers
0x4000A800 - 0x4000ABFF	CAN CCU		Section 56.5: FDCAN registers
0x4000A400 - 0x4000A7FF	FDCAN2		Section 56.5: FDCAN registers
0x4000A000 - 0x4000A3FF	FDCAN1		Section 56.5: FDCAN registers
0x40009400 - 0x400097FF	MDIOS		Section 54.4: MDIOS registers
0x40009000 - 0x400093FF	OPAMP		Section 29.6: OPAMP registers
0x40008800 - 0x40008BFF	SWPMI		Section 53.6: SWPMI registers
0x40008400 - 0x400087FF	CRS		Section 9.8: CRS registers
0x40007C00 - 0x40007FFF	UART8		Section 48.7: USART registers
0x40007800 - 0x40007BFF	UART7		Section 48.7: USART registers
0x40007400 - 0x400077FF	DAC1		Section 26.7: DAC registers
0x40006C00 - 0x40006FFF	HDMI-CEC		Section 59.7: HDMI-CEC registers
0x40005C00 - 0x40005FFF	I2C3		Section 47.7: I2C registers
0x40005800 - 0x40005BFF	I2C2		Section 47.7: I2C registers
0x40005400 - 0x400057FF	I2C1		Section 47.7: I2C registers
0x40005000 - 0x400053FF	UART5		Section 48.7: USART registers
0x40004C00 - 0x40004FFF	UART4		Section 48.7: USART registers
0x40004800 - 0x40004BFF	USART3		Section 48.7: USART registers
0x40004400 - 0x400047FF	USART2		Section 48.7: USART registers
0x40004000 - 0x400043FF	SPDIFRX1		Section 52.5: SPDIFRX interface registers
0x40003C00 - 0x40003FFF	SPI3 / I2S3		Section 50.11: SPI/I2S registers
0x40003800 - 0x40003BFF	SPI2 / I2S2		Section 50.11: SPI/I2S registers
0x40002C00 - 0x40002FFF	Reserved		Reserved
0x40002400 - 0x400027FF	LPTIM1		Section 43.7: LPTIM registers
0x40002000 - 0x400023FF	TIM14		Section 39.4: TIM2/TIM3/TIM4/TIM5 registers
0x40001C00 - 0x40001FFF	TIM13		Section 39.4: TIM2/TIM3/TIM4/TIM5 registers
0x40001800 - 0x40001BFF	TIM12		Section 39.4: TIM2/TIM3/TIM4/TIM5 registers
0x40001400 - 0x400017FF	TIM7		Section 42.4: TIM6/TIM7 registers
0x40001000 - 0x400013FF	TIM6		Section 42.4: TIM6/TIM7 registers
0x40000C00 - 0x40000FFF	TIM5		Section 39.4: TIM2/TIM3/TIM4/TIM5 registers
0x40000800 - 0x40000BFF	TIM4		Section 39.4: TIM2/TIM3/TIM4/TIM5 registers
0x40000400 - 0x400007FF	TIM3		Section 39.4: TIM2/TIM3/TIM4/TIM5 registers
0x40000000 - 0x400003FF	TIM2		Section 39.4: TIM2/TIM3/TIM4/TIM5 registers

1. Accessing a reserved area results in a bus error. Accessing undefined memory space in a peripheral returns zeros.